

Schiff base modified polyurethane coating with enhanced mechanical property for marine antifouling application

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ABSTRACT

Silicone coatings cannot completely prevent the adhesion of marine organisms on hull surfaces under static conditions. In this study, a Schiff base modified polyurethane antifouling coating (PUx-SBSi) was prepared by a facile method. The silicone modified by Schiff base not only retained the low surface energy characteristic, but also effectively addressed the inherent issue of insufficient static antifouling performance in single silicone-based materials. The PUx-SBSi coatings inhibited the adhesion of marine bacteria (*Pseudomonas xiamenensis* (*P. xiamenensis*)) up to 95.3 %. Furthermore, the PUx-SBSi coatings exhibited exceptional mechanical property and a significantly higher adhesion strength, with a maximum of 7.3 MPa, which overcome the limitations of inadequate mechanical property and weak adhesion strength of single silicone-based coatings. Therefore, PUx-SBSi coatings with both mechanical properties and static antifouling performance are anticipated to effectively protect ship surfaces from marine organisms, playing a crucial role in the sustainable development of functional coating materials.

1. Introduction

Marine biological fouling, primarily caused by microorganisms, marine animals, and plants in the ocean, adheres to and proliferates on the surfaces of hulls, marine facilities, or weapon systems [1–3]. Applying antifouling coatings to equipment surfaces is a direct and effective way to mitigate marine biofouling [4–6]. Traditional antifouling coatings combat biofouling by gradually releasing toxic fungicides into seawater during operation leading to significant environmental pollution and harm to marine ecosystems and aquatic life [7–9]. Furthermore, certain antifouling coatings exhibit inadequate durability or suboptimal performance in practical settings, necessitating frequent maintenance and recoating [10,11]. To address these challenges, researchers are developing eco-friendly antifouling coatings designed to minimize or eliminate detrimental environmental impacts while preserving effective antifouling properties [12,13].

Polyurethanes are widely used in antifouling coatings due to their excellent mechanical and other beneficial properties [14,15], the

limited antifouling activity of polyurethanes limits their application. To address this disadvantage, polyurethane is usually blended with other resins. Among them, silicon materials with low surface energy characteristic are often prepared as fouling release coatings (FRCs), and the attached organisms on their surfaces are easy to be removed under water current flushing [16–18]. Silicone-modified polyurethane coatings alter the molecular interaction between marine fouling organisms and the antifouling coating, reducing interfacial adhesion and preventing their attachment to the surface [19,20]. Galhenage et al. investigated the impact of incorporating various types and concentrations of silicone on the performance of siloxane-polyurethane systems [21]. Both laboratory and field tests demonstrated that adding silicone could reduce the minimum adhesion strength of large fouling organisms. Zheng et al. developed a robust microstructure of low-surface energy polyurethane-fluorinated polysiloxane (PU-^FPDMS) with improved mechanical properties, surface energy, and bond strength [22]. However, silicone modified polyurethane coatings exhibit inadequate antifouling performance under static conditions (e.g., when ships are docked), restricting

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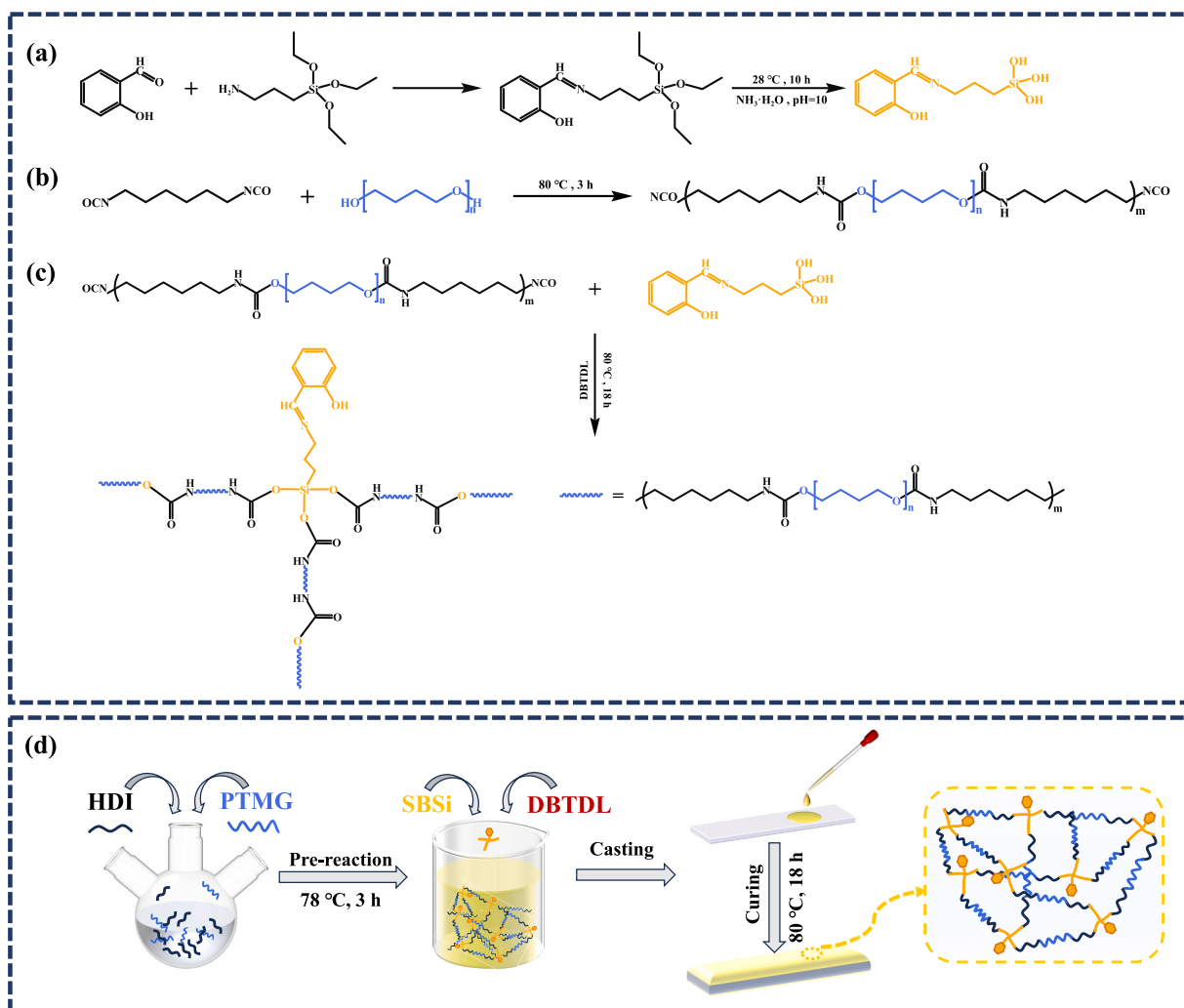
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Scheme 1. (a) Synthesis of Schiff base-containing triethoxysilane (SBSi). (b) Synthesis of Polyurethane prepolymer (PUx). (c) Preparation of the PUx-SBSi coatings. (d) Flow diagram for the preparation of the PUx-SBSi coatings.

their broader application.

In addressing the above issue, functional groups with antifouling and antibacterial properties are typically incorporated into the coating through grafting [23,24]. Schiff bases, known for their exceptional antibacterial activity, can be synthesized under mild conditions through a straightforward condensation reaction involving aldehydes or ketones and amines [25,26]. Interaction of Schiff base compounds with the phospholipid bilayer in bacterial cell membranes results in the disruption of membrane integrity, leading to bacterial death [27,28]. Based on their morphological properties, Schiff base antibacterial antifouling material can be divided into Schiff base antibacterial antifouling agents and poly-Schiff base antibacterial antifouling block materials/coatings [29]. The degradable and self-renewing surfaces of poly-Schiff base materials offer excellent antifouling properties and represent a good strategic direction for addressing the urgent need for marine antifouling. However, the majority of currently available biodegradable poly-Schiff base antifouling materials are primarily derived from synthetic chemicals, which continue to present a potential risk to ecosystems. Thus, the integration of Schiff base, known for its bactericidal properties, with silicone through a straightforward condensation reaction can enhance static antifouling performance without causing environmental pollution.

In this work, a Schiff base modified polyurethane (PUx-SBSi) antifouling coating with outstanding mechanical properties is prepared. Schiff base modified silane (SBSi) is synthesized as a curing agent

through a straightforward condensation reaction. The SBSi effectively diminishes the interaction between marine fouling organisms and the coating surface. This modification endows the coating with exceptional static antifouling properties, thereby inhibiting the attachment of barnacles, mussels, and other biofouling species. Meanwhile, the coatings demonstrated formidable mechanical strength and interfacial adhesion, which is attributed to hydrogen bonding in the polyurethane, covalent cross-linking sites provided by SBSi, and rigid benzene ring complexation. This study provides an innovative and practical method for preparing marine antifouling coatings with both outstanding antifouling and mechanical properties.

2. Experimental section

2.1. Materials

3-Aminopropyltriethoxysilane (AR, 99 %) and ammonium hydroxide (AR, 99 %) were bought from Shanghai Aladdin Biochemical Technology Co., Ltd. Salicylaldehyde (AR, 98 %) and hexamethylene diisocyanate (HDI; AR, 99 %) were provided by Shanghai Maclean Biochemical Co., Ltd. Poly (tetramethylene ether glycol) (PTMG; Mn = 2000 g/mol) was acquired from Jining Huakai Resin Co., Ltd. *N,N*-Dimethylformamide (DMF, AR) was purchased from Tianjin Guangfu Technology Development Co., Ltd. Ethanol absolute (AR) was bought from Tianli Chemical Reagent Co., Ltd. Sylgard 184 (PDMS, Dow Corning.) were used as

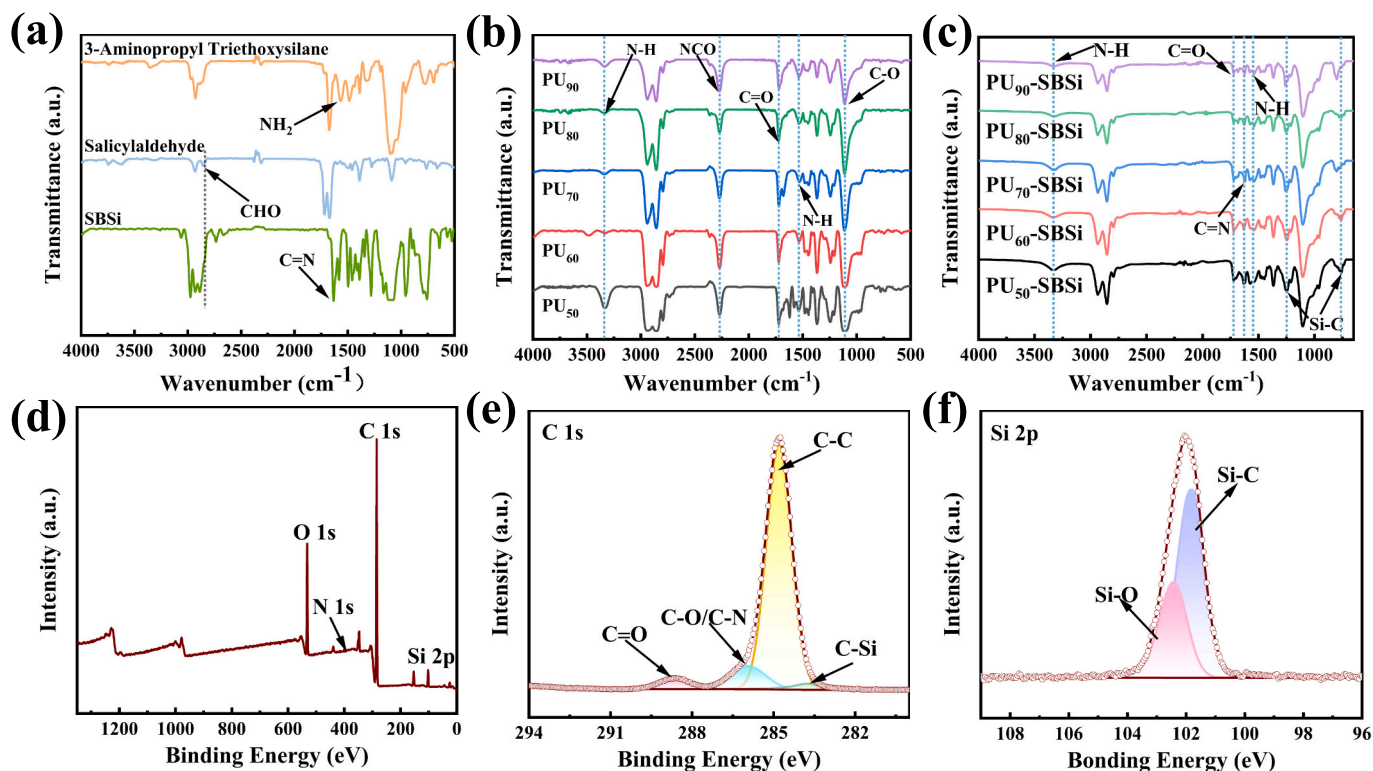


Fig. 1. (a) The FT-IR spectra of SBSi. (b) The FT-IR spectra of PU_x prepolymers. (c) The ATR-FTIR spectra of the PU_x-SBSi coatings. (d) The XPS spectrum of PU₆₀-SBSi coating and corresponding high-resolution spectra of (e) C 1s (f) Si 2p.

received. Sylgard 184 was applied in a 10:1 ratio (w/w) of siloxane and curing agent.

2.2. Synthesis of Schiff base-containing triethoxysilane (SBSi)

According to Scheme 1(a) 25.64 g (210 mmol) salicylaldehyde was dissolved in 80 mL of ethanol (named as solution A). 46.5 g (210 mmol) 3-aminopropyltriethoxysilane was taken and added dropwise to solution A over 2 h. The reaction was heated to 80 °C under N₂ atmosphere for 5 h. Ethanol was removed from the product using a rotary evaporator, yielding a yellow liquid identified as the precursor Schiff base-containing triethoxysilane (SBSi).

After that, a certain amount of ethanol was added to deionized water, placed on a magnetic stirrer, stirred well, and then SBSi was slowly dropped. The mass ratio of SBSi, deionized water, and ethanol was 1:1:6. The pH was adjusted to 10 by dropwise addition of ammonia solution and stirring for 6 h at room temperature until the solution became clear. After the reaction, the solvent was removed by rotary evaporator. The yield of SBSi was calculated to be 91.4 % according to Eq. S4.

2.3. Synthesis of polyurethane prepolymer (PU_x)

Polyurethane prepolymer (PU_x) was synthesized through a stepwise polymerization method, utilizing 14.8 g (88 mmol) HDI and 60 g (30 mmol) PTMG (Scheme 1(b)). In order to avoid the moisture in PTMG to react with HDI and explosive polymerization, the PTMG was heated and vacuumed to eliminate the residual moisture before adding HDI. After cooling to 40 °C, HDI was introduced dropwise into the flask with a stirrer, condenser tubes, and thermometer. The reaction was maintained at 80 °C for 3 h. The obtained samples were named after PU₆₀. Similarly, PU₉₀, PU₈₀, PU₇₀, and PU₅₀ were obtained under the same reaction conditions using 90 g (45 mmol), 80 g (40 mmol), 70 g (35 mmol), and 50 g (25 mmol) of PTMG, respectively.

2.4. Preparation of PU_x-SBSi coatings

Typically, PU_x, SBSi and catalyst dibutyltin dilaurate (DBTDL, 15 mg) were dissolved in DMF and stirred with a magnetic bar for 1 h (Scheme 1(c)). SBSi was added as a curing agent, and the added mass was calculated by the following formula (1):

$$m = \frac{m_{\text{PU}_x} \times \text{NCO}\% \times M}{R \times 42 \times n} \quad (1)$$

Here, m and m_{PU_x} stand for the amount of SBSi added and the mass of PU_x prepolymer, respectively. NCO% denotes the isocyanate content of PU_x prepolymer. M stands for the molar mass of SBSi, and R is the chain expansion ratio ($R = 1.02$), the constant 42 corresponds to the molecular weight of the isocyanate group, and n is the functionality of SBSi ($n = 3$). The specific formulations are provided in Table S1.

The resulting mixture was either applied to glass slides or poured into polytetrafluoroethylene (PTFE) moulds. Bubbles were subsequently removed in a vacuum oven at 50 °C under negative pressure. The glass slides and PTFE moulds were first immersed in deionized water (DI) for 10 min. The surface of the substrate was wiped with cotton moistened with ethanol to remove dust impurities, etc., and finally placed in a clean and dust-free environment at room temperature to dry before use. The coatings were cured in a blast drying oven at 80 °C for 18 h. Following a 5–7 days period at room temperature, a series of tests were conducted. The coating is designated as PU_x-SBSi, where x indicates the amount of PTMG in the polyurethane prepolymer.

2.5. Characterization

Fourier transform infrared spectrometer (FTIR) spectroscopy was carried out by an FTIR spectrophotometer (Spectrum 100, PerkinElmer, USA) range of 4000–500 cm⁻¹. The Attenuated Total Reflection-Fourier Transform Infrared (ATR-FTIR) spectra were obtained on Bruker Tensor 27 spectrometer. The X-ray photoelectron spectroscopy (XPS) was

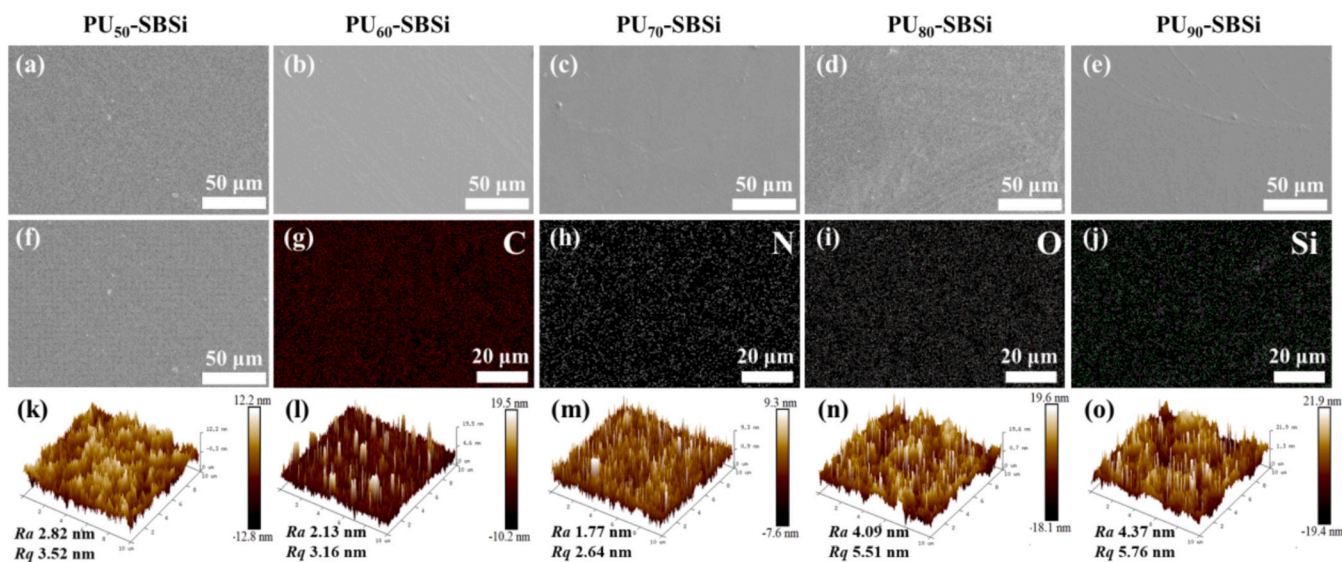


Fig. 2. The surface topography and mapping images of the coatings. (a–e) The SEM images of the PU_x-SBSi coatings. (f–j) The SEM image and mapping of C, N, O, and Si elements for the PU₅₀-SBSi coating. (k–o) The 3D morphology images of AFM of the PU_x-SBSi coatings.

measured using a surface analysis system (Thermofisher Escalab Xi+). The surface topography of the coatings was characterized using the Apreo S scanning electron microscope (SEM) of the FEI Corporation of America, and the coatings were coated on $5 \times 5 \text{ mm}^2$ copper sheets. The surface profile was tested using an atomic force microscope (AFM, Dimension Icon) with a test range was $10 \times 10 \text{ μm}^2$.

The details and other characterization were listed in Supplementary material.

3. Results and discussion

3.1. Structural and morphological characterization of the PU_x-SBSi coatings

The FT-IR spectrum of salicylaldehyde, 3-Aminopropyltriethoxysilane and SBSi presented in Fig. 1(a) was analyzed to characterize its chemical structure. In the spectra of salicylaldehyde and 3-aminopropyltriethoxysilane, the absorption bands around 2750 cm^{-1} and 2850 cm^{-1} correspond to the characteristic stretching vibrations of the -CHO group in salicylaldehyde. Meanwhile, the absorption near 1570 cm^{-1} is attributed to the bending vibration of the -NH₂ group in 3-aminopropyltriethoxysilane [30]. The absorption around 1640 cm^{-1} is attributed to the characteristic C=N stretching band in SBSi, indicating the complete removal of the -CHO and -NH₂ groups [31]. The ¹H NMR analysis was conducted to examine the molecular structure of SBSi and PU_x (Figs. S1 and S2). Compared with the spectrum of salicylaldehyde, the peak of δ 10.25 (s, 1H, -CHO) of aldehyde group disappeared and the peak of imine (δ 8.54) appeared, which indicated that the aldehyde group reacted with the amino group to form an imine bond, and therefore, it could be determined that SBSi was synthesized successfully. Additionally, the chemical shifts at 3.45 ppm and 1.51 ppm are attributed to the -CH₂ group. The peak corresponding to the hydroxyl group disappears completely at 4.70 ppm, indicating that the terminal hydroxyl group is fully reacted, while the peak appearing near 8.01 ppm corresponds to the peak of the acetamide, indicating the generation of the urethane structure [32]. Thus, based on the FT-IR spectroscopy and ¹H NMR results, the successful preparation of SBSi is confirmed.

The FT-IR spectra of PU_x prepolymers are shown in Fig. 1(b). The infrared spectra of the PU_x prepolymer exhibit peaks at 1534 cm^{-1} and 3350 cm^{-1} , corresponding to the bending and stretching vibrations of -N-H characteristic functional group. The characteristic peak around 2270 cm^{-1} is generated by the bending vibration of the -NCO group of

HDI. Furthermore, the peaks observed around 1718 cm^{-1} are generated by the vibrations of the C=O bonds in -COO- and -NHCOO- group [33,34]. These findings demonstrate that PU_x prepolymers were successfully prepared. As illustrated in Fig. 1(c), it can be analyzed that two typical absorption peaks of Si-C group, the stretching vibration and the variable angle vibration absorption peak, appear at 1250 cm^{-1} and 795 cm^{-1} , respectively, indicating that SBSi has been successfully introduced into the coatings. Moreover, the appearance of the C=N absorption peak in the SBSi structure at 1640 cm^{-1} , along with the absence of the absorption peak at 2270 cm^{-1} in the spectra of all the PU_x-SBSi coatings, confirms that the -NCO group in the PU_x-SBSi coatings were fully involved in the reaction and the coatings were fully cured [35,36].

Fig. 1(d) presents the wide-scan XPS spectrum of the PU₆₀-SBSi coating and the characteristic signals of C 1s (284.8 eV), N 1s (397 eV), O 1s (532 eV), and Si 2p (102 eV) were observed on PU₆₀-SBSi surface. The high-resolution spectrum of C 1s is illustrated in Fig. 1(e), which could be fitted into the peaks located at C=O (288.6 eV), C-N/C-O (285.9 eV), C-C (284.8 eV), and C-Si (283.7 eV) peaks, respectively. The spectrum of Si 2p can be divided into two peaks centered at 102.44 eV and 101.85 eV which belong to the binding energies of Si-O and Si-C, respectively [37]. The N 1s high-resolution spectrum of the coating has two peaks located in the range of $398\text{--}400 \text{ eV}$ (Fig. S3). One of the peaks belongs to C=N bond from SBSi. Further, wherein the -N-H group arises due to the reaction of the -NCO group with the -OH group, thereby further confirming the presence of the -NHCOO- group and the disappearance of the -NCO group. These facts indicate that the SBSi component is successfully participated in polymerization.

As shown in Fig. 2(a–j), the coatings all exhibit smooth surfaces, proving that the PU_x-SBSi coatings are homogeneous polymers and successfully prepared. The analysis reveals an even distribution of C, N, O, and Si elements on the surface of the PU_x-SBSi coatings. In addition, the specific elemental content of the PU_x-SBSi coatings is shown in Table S2. As the proportion of SBSi increases, the proportion of silicon atoms on the coating surface also increases. The roughness of the coating surface is investigated by AFM, as shown in Fig. 2(k–o). As the SBSi content increases, resulting in a gradual decrease in the roughness of the coating. When the SBSi content increases further, the introduction of a large number of small molecules increases the microphase separation of the coating, resulting in a small increase in the roughness of the coating. In general, silicone-modified urethane coatings typically have smooth surfaces and low microphase separation when silicone loading is relatively low ($<10 \text{ wt\%}$) [38]. Since the variation in coating roughness

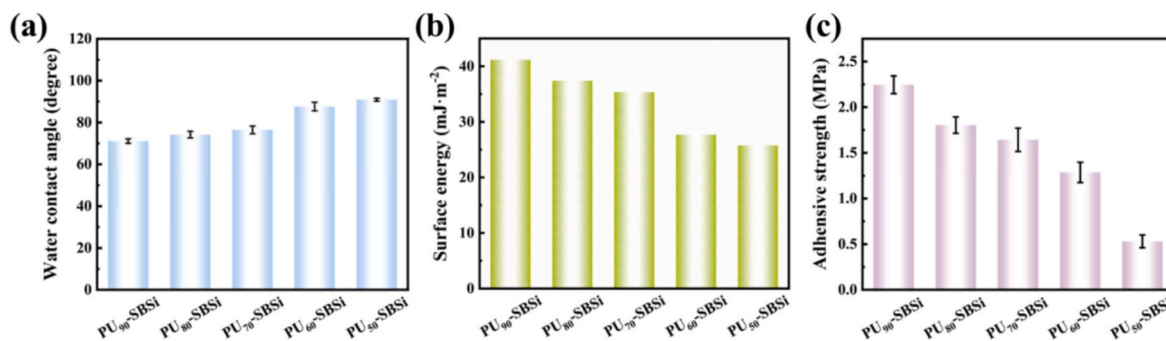


Fig. 3. (a) The water contact angle of the PU_x-SBSi coatings. (b) Surface energy of the PU_x-SBSi coatings. (c) The removal strength of pseudo-barnacles of the PU_x-SBSi coatings. Each sample was repeated at least three times, and the results were averaged.

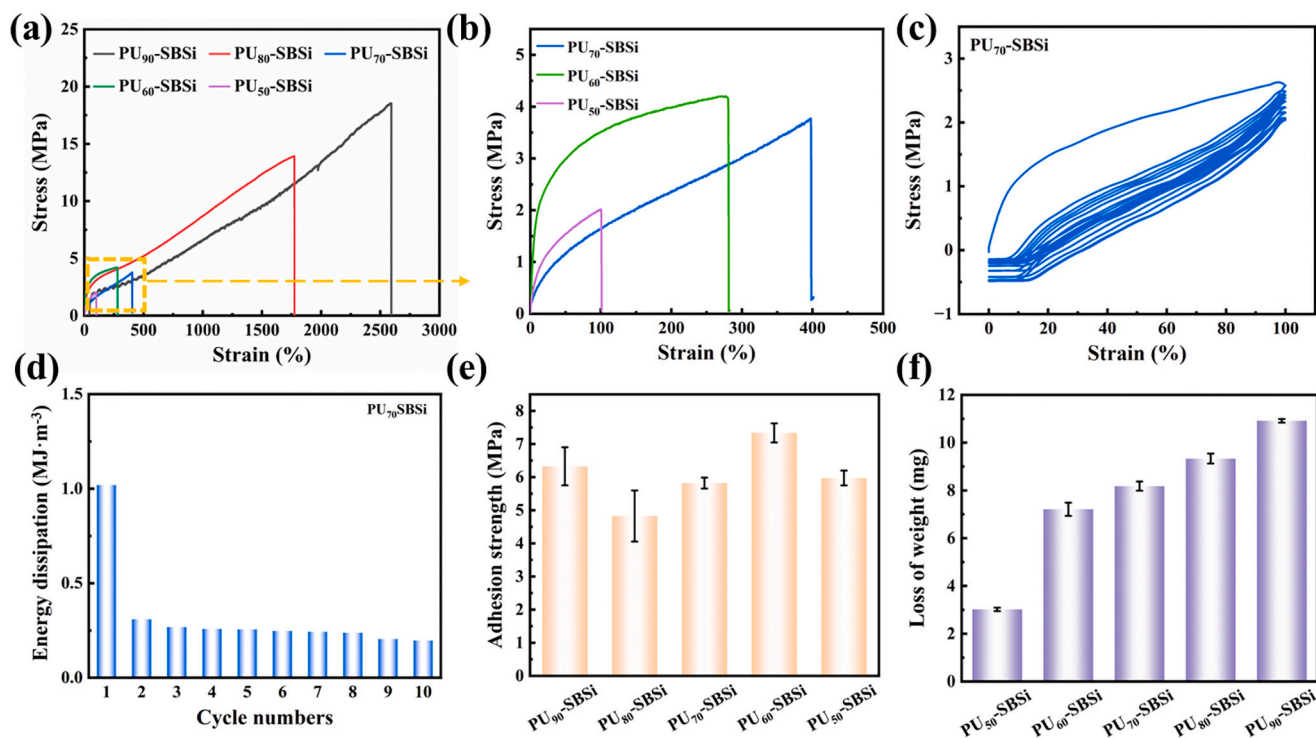


Fig. 4. (a) Graph of stress-strain of PU_x-SBSi coatings. (b) The partial enlargement of panel (a). (c) The loading-unloading curve and (d) dissipative energy of PU₇₀-SBSi coating under a constant strain of 100 % for 10 cycles. (e) The adhesion strength of PU_x-SBSi coatings. (f) The coatings mass reduction following the abrasion test.

consistently remained within the nanometer range, it was not discernible under SEM magnification. Therefore, the SEM images show a flat surface of the coatings without cracks or wrinkles.

3.2. Surface property of PU_x-SBSi coatings

The surface wettability of the coatings was examined by measuring the water contact angle (WCA). The average static WCA of the PU_x-SBSi coatings (Fig. 3(a)) were measured to be $71.1 \pm 1.0^\circ$, $72.3 \pm 1.5^\circ$, $76.4 \pm 1.8^\circ$, $87.5 \pm 2.0^\circ$, and $90.8 \pm 0.6^\circ$, respectively. As the SBSi content increases, the WCA of the PU_x-SBSi coatings shows a gradual increase, and the WCA of the PU₅₀-SBSi coating reaches $90.8 \pm 0.6^\circ$. This is attributed to the hydrophobicity of the benzene ring in SBSi. The value of surface energy (SE) of the coating is an important factor affecting the bond strength of the coating surface. Therefore, CA test is performed at non-polar CH₂I₂ (Fig. S4) and the SE of the PU_x-SBSi coatings was calculated. The SE of the coating with the highest amount of SBSi added was $25 \text{ mJ}\cdot\text{m}^{-2}$, while the SE of the coating with the lowest amount of

SBSi added is $42 \text{ mJ}\cdot\text{m}^{-2}$ (Fig. 3(b)). The incorporation of SBSi leads to a reduction in the SE of the coatings. Meanwhile, the SE values of PU₅₀-SBSi and PU₆₀-SBSi coatings are between $25 \text{ mJ}\cdot\text{m}^{-2}$ and $30 \text{ mJ}\cdot\text{m}^{-2}$, which have the property of making it less likely for fouling organisms to adhere to the surface [1]. The PU_x-SBSi coatings were applied to glass slides to investigate the sliding of various liquid contaminants (water, orange juice and coffee) on their surfaces (whether or not they left traces, the angle of sliding, etc.) and to compare the outcomes with the results of tests on blank glass substrates without coatings (Fig. S5). These different liquid contaminants can be easily removed from the coating surface. This removal occurs through sliding-off without leaving any traces behind. The PU₅₀-SBSi coating, which had the lowest SE, had excellent antifouling performance, with DI water sliding off the surface quickly at a surface inclination of about 35° . The analysis of the results from the pseudo-barnacle experiment presented in Fig. 3(c) demonstrates a clear correlation between the force required to remove pseudo-barnacles from the coating surface and the SBSi content. Specifically, as the SBSi content increases, there is a corresponding decrease in the force

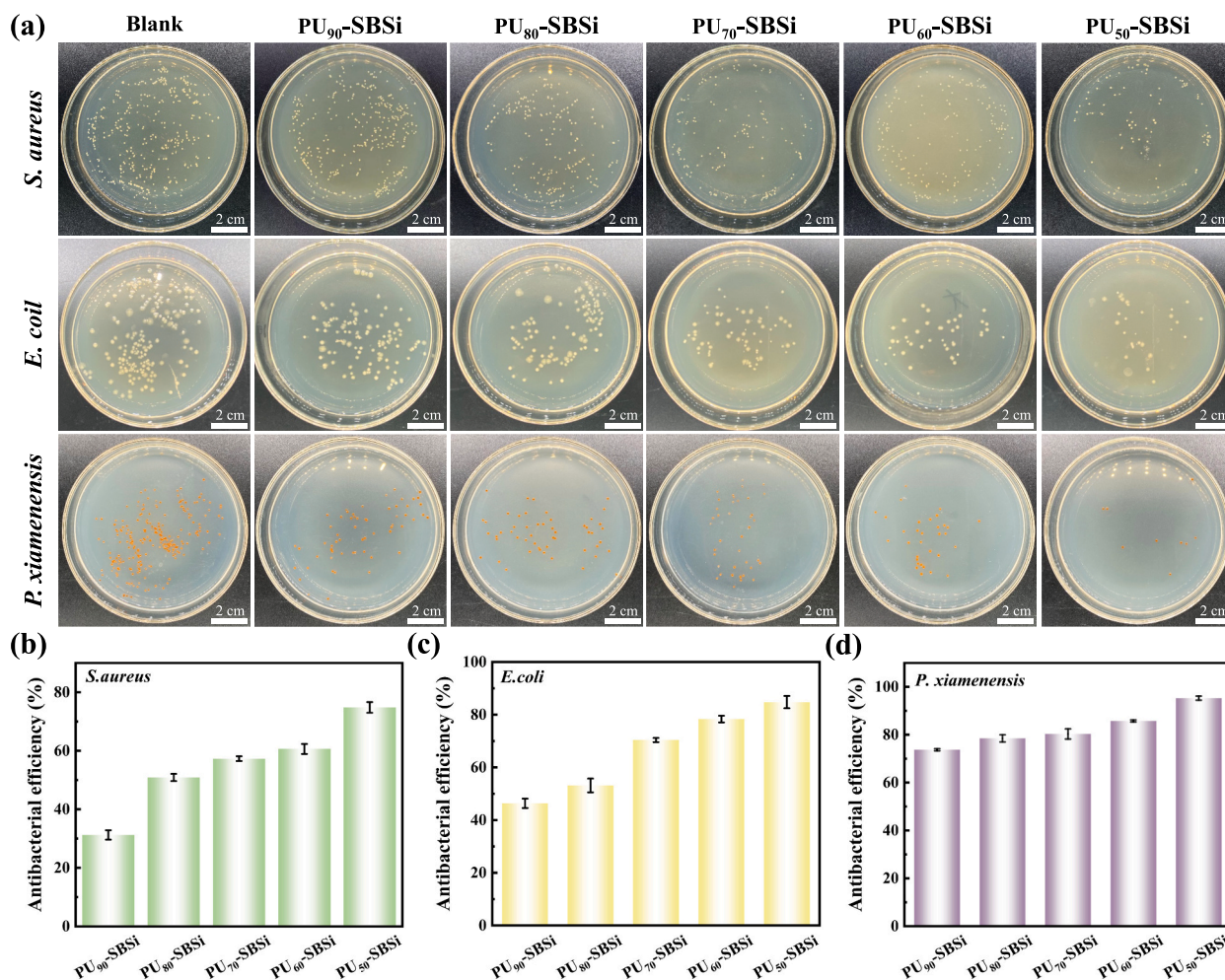


Fig. 5. (a) Optical photographs of *S. aureus*, *E. coli* and *P. xiamenensis* colonies after co-culture with PUx-SBSi coatings. The antibacterial efficiency of (b) *S. aureus*, (c) *E. coli* and (d) *P. xiamenensis* colony on PUx-SBSi coatings.

necessary to detach the pseudo-barnacles from the coating surface [39]. Notably, the minimum strength required to remove the pseudo-barnacles bonded to the surface of the PU₅₀-SBSi coating is about 0.53 MPa. This result is consistent with previous SE test.

3.3. Mechanical properties of PUx-SBSi coatings

Since the coatings on the surface of the ship will be continuously eroded by the sea water in the course of sailing, it is necessary to test the mechanical durability of the coatings. The mechanical properties of PUx-SBSi coatings were evaluated through a tensile test, as shown in Fig. 4(a) and (b). It is observed that the tensile strength of the coatings decreases as the SBSi content increases. Notably, the highest strength is recorded of PU₉₀-SBSi, while the strength of PU₈₀-SBSi shows a declining trend. This decrease in strength were attributed to the heightened cross-linking density, which hinders the movement of molecular chains, reducing toughness [40]. Furthermore, the mechanical properties of the PUx-SBSi coatings were further evaluated using cyclic tensile tests. During the 10 loading-unloading cyclic experiments, the cyclic tensile speed was kept at 25 mm/min and the tensile strain was set at 100 %. The dissipated energy of the coating was calculated by integrating over the area included in the cyclic stretching curve. As shown in Figs. 4(c–d) and S6, the obtained results further demonstrate the mechanical durability of the PUx-SBSi coatings.

The outcomes of the adhesion test of the coatings applied are presented in Fig. 4(e) and demonstrate their excellent adhesion properties,

with all coatings exhibiting adhesion strengths above 4.82 ± 0.77 MPa. The adhesion strength is attributed to the presence of hydrogen bonds from the urethane groups and polar bonds from PTMG. To assess abrasion resistance, the coatings underwent testing using an abrasion tester, with identical testing conditions for all PUx-SBSi coatings. The mass of the coatings before and after abrasion was meticulously measured using an analytical balance to determine the mass loss and evaluate the impact of SBSi addition on abrasion resistance. The result in Fig. 4(f) reveals that the abrasion amount of the coating gradually increases with the addition of SBSi content. PU₅₀-SBSi exhibited the lowest abrasion amount at only 3.02 ± 0.072 mg. As the SBSi content increases, the crosslinking density of the coating increases, the increase in benzene rings, and the hydrogen bonding leads to an increase in the abrasion resistance of the coating. As shown in Fig. S7, according to multiple Gaussian fit, three independent fit peaks in PU₇₀SBSi were observed at about 1721, 1700, and 1682 cm^{-1} to estimate the phase separation, corresponding to free C=O groups in hard segments, hydrogen bonded C=O groups in the soft segments and the disordered hydrogen bonded C=O groups in hard segments, respectively. By analyzing the proportion of various types of hydrogen bonds in the spectrum, it can be seen that 54.25 % of them are ordered hydrogen bonds. Since the strength of hydrogen bonds is lower compared to chemical bonds, the energy of hydrogen bonds is released first [41]. In addition, Table S3 shows the adhesion and tensile strengths of the silicone-modified coatings in comparison with other literatures, indicating the excellent mechanical properties of the prepared PUx-SBSi coatings. From Fig. S8, it can be

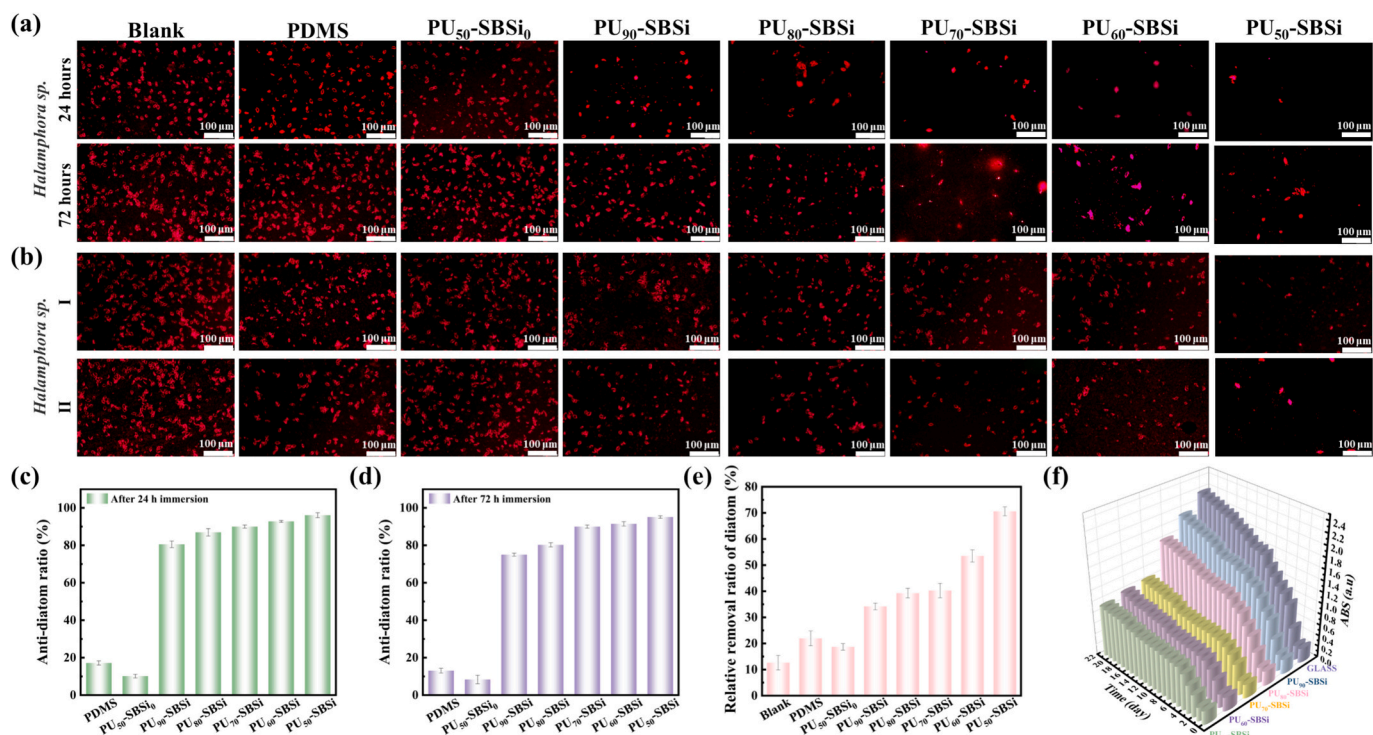


Fig. 6. (a) Representative fluorescence microscopy images of PDMS, PU₅₀-SBSi₀, and PU_x-SBSi coatings immersed in *Halamphora* sp. solutions for 24 h and 72 h, with a scale bar of 100 μ m. (b) The PDMS, PU₅₀-SBSi₀, and PU_x-SBSi coatings following a week of immersion in *Halamphora* sp. solution, the coating surface was washing with a 20 Pa water stream. (I) and (II) are the microscopy images of diatoms attached to the coating surface before and after washing. The scale is 100 μ m. The resistance ratio of PDMS, PU₅₀-SBSi₀, and PU_x-SBSi coatings after (c) 24 h and (d) 72 h immersion in *Halamphora* sp. solution. (e) Diatom release ratio of PDMS, PU₅₀-SBSi₀, and PU_x-SBSi coatings under dynamic condition. (f) The *Nitzschia closterium* growth curve of the coatings for 21 days.

observed that the two samples exhibited no significant mass change between 0 and 200 °C. As the temperature increased, a more pronounced weight loss was noted at 350 °C, and by 450 °C, the coatings had nearly completed their combustion decomposition. This indicates that the coating demonstrates better thermal stability.

3.4. Antifouling performance test on PU_x-SBSi coatings

The anti-biofouling performance of the PU_x-SBSi coatings was examined using bacteria, diatoms, and mussels as the typical fouling organisms. To assess the ability of the PU_x-SBSi coating to inhibit bacterial adhesion, the antibacterial assays were conducted using three representative bacteria: *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*), which are commonly found in daily environment [42], and the *Pseudomonas xiamenensis* (*P. xiamenensis*) to demonstrate the antibacterial capacity of the coating in the marine environment [43]. The optical photographs of the colonies of *S. aureus*, *E. coli*, and *P. xiamenensis* are shown in Fig. 5(a). From Fig. 5(b–d), the PU₅₀-SBSi coating showed the best antibacterial ability with 74.8 ± 1.8 % against *E. coli*, 84.8 ± 2.3 % against *S. aureus* and 95.3 ± 0.8 % against *P. xiamenensis*. Notably, the antibacterial performance of these coatings is found to be positively correlated with the SBSi content, indicating its significant enhancement in preventing bacterial adhesion. In order to investigate the antifouling mechanism of the coatings, two silicone-based coatings were prepared, including commercial PDMS and PU₅₀-SBSi₀ (hydrolyzed 3-aminopropyltriethoxysilane grafted into prepolymer PU₅₀). The inhibition efficiency of *P. xiamenensis* was 19.5 ± 2.4 % and 11.2 ± 0.72 % for PDMS and PU₅₀-SBSi₀, respectively (Fig. S9). The experimental results show that the antifouling ability of the coating mainly comes from the Schiff base structure in SBSi [44].

Halamphora sp. (benthic diatom) was chosen as representative species to assess the anti-diatom adhesion property of PU_x-SBSi coating. The coatings were placed in artificial seawater where *Halamphora* sp.

were cultivated for several days to observe the growth of it on the surface of the coatings. The anti-diatom ratio of PU_x-SBSi coating was calculated by comparing it to a control group of blank glass. The results presented in Fig. 6 illustrate that the coating effectively inhibits adhesion for both types of diatoms. Notably, the adhesion inhibition effect improved with increasing amounts of SBSi. The calculation of the number of diatoms on the surface showed that the anti-diatom ratios of PU_x-SBSi against *Halamphora* sp. after 24 h of immersion are 78.2 ± 1.7 %, 88.1 ± 1.0 %, 91.2 ± 0.2 %, 94.2 ± 0.5 % and 95.8 ± 0.3 %, respectively. Relatively, the inhibitory ratios after 72 h of immersion are 70.3 ± 0.7 %, 87.5 ± 1.6 %, 88.5 ± 1.5 %, 93.1 ± 1.0 % and 94.7 ± 0.6 %, respectively (Fig. 6(c)). In addition to the diatom resistance test under static conditions, the fouling-release of the coatings after washing with high pressure water is a key test for the antifouling performance of PU_x-SBSi coatings. Fig. 6(b) shows the changes in the number of diatoms adhered to the PU_x-SBSi coatings surface before and after rinsing with 20 Pa water stream. The removal of diatoms from the PU_x-SBSi coatings exhibited a gradual increase in correlation with the rising SBSi content. (Fig. 6(d)), and the removal of diatoms of PU₅₀-SBSi was as high as 76.6 ± 4.5 %. This observation indicates that with increasing SBSi content, the benzene ring concentration rises, leading to enhanced surface hydrophobicity and reduced surface energy, which collectively promote the release of fouling organisms. Fig. 6(e) illustrates the growth of *Nitzschia closterium* (planktonic diatom) over a 21-day period during which the coating was immersed in diatom solution. The diatom absorbance values of PU₅₀-SBSi, PU₆₀-SBSi, and PU₇₀-SBSi solutions are significantly lower than those of PU₈₀-SBSi and PU₉₀-SBSi. In addition, the diatom release ratio of PDMS and PU₅₀-SBSi₀ coatings after 3 days of immersion in the diatom solution were 13.1 ± 1.3 % and 8.3 ± 2.2 %, respectively. Although PDMS and PU₅₀-SBSi₀ had some fouling release capability, the adhesion inhibition of diatom was not as efficient as that of the PU_x-SBSi coatings. This further confirms that the antifouling ability of the coatings mainly originates from the Schiff base structure

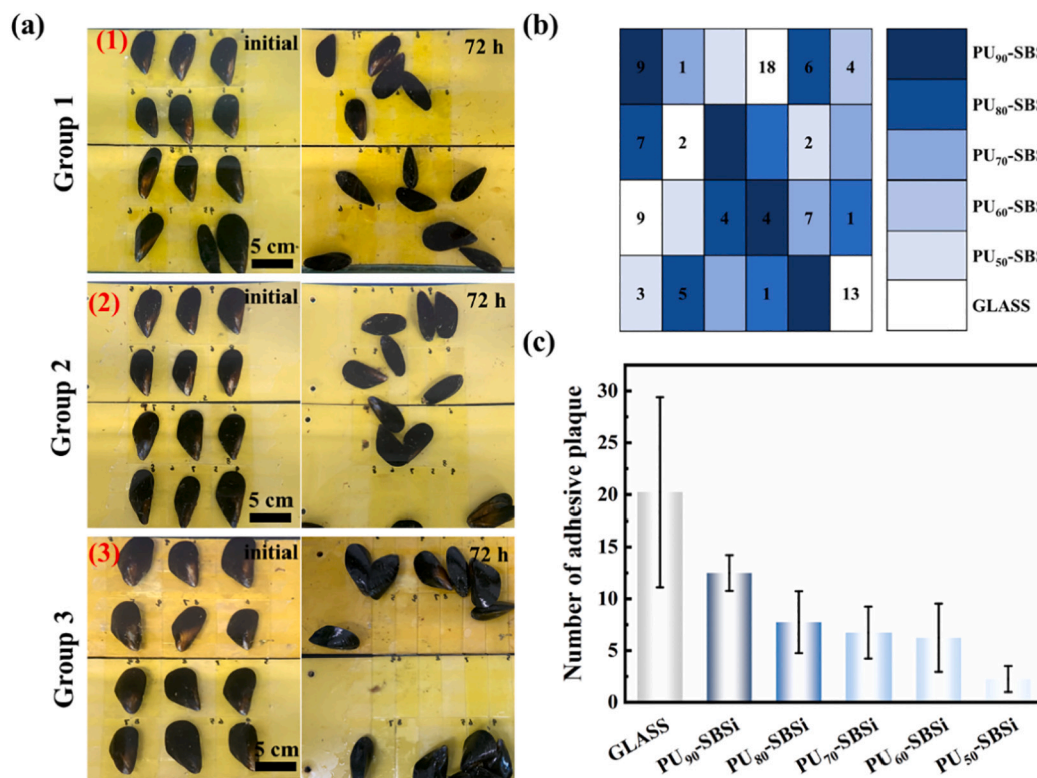


Fig. 7. (a) Photos of the mussels in three repeated mussel multi-choice settlement experiments. (b) A scheme representation of the first group of mussel multi-choice settlement experiments, each number indicates the number of mussel plaques attached to the different coatings after 72 h. (c) The number of plaques secreted by mussels on glass slides and PU_x-SBSi coatings.

[45].

Mussels, as larger fouling organisms, generally cause more significant damage than diatoms. Therefore, we further investigated the antifouling performance of the PU_x-SBSi coatings against macro-organisms [46]. At the start of the experiment, the twelve adult marine mussels were evenly distributed on the epoxy plate, where different coatings were assembled together. Three sets of tests were performed to evaluate the influence of different PU_x-SBSi coatings on the selective attachment behavior of mussels. As shown in Figs. 7 and S10, at the beginning of the experiment the mussels were searching for a suitable adhesion surface, and after 72 h most of them moved to epoxy plate, glass slides or adhered with other mussels. By counting the number of mussel plaques on the test board, it was demonstrated that the PU_x-SBSi coatings have excellent anti-mussel adhesion properties. Overall, the mussel selection experiments reveal a clear tendency for mussels to avoid PU_x-SBSi coatings with higher SBSi content.

4. Conclusion

In conclusion, a Schiff base modified polyurethane antifouling coating (PU_x-SBSi) was prepared, combining the exceptional fouling release properties of silicone with the outstanding mechanical strength of polyurethane. The Schiff base structure derived from SBSi has the ability to inhibit fouling adhesion at different scales, while the benzene ring structure provides fouling release. PU_x-SBSi coatings with anti-fouling additives show significant antibacterial properties and excellent resistance to diatoms and mussels. Additionally, the coatings demonstrate exceptional mechanical properties alongside superior static anti-fouling performance, surpassing those of most previously reported PDMS coatings, which suffer from inadequate substrate adhesion and mechanical durability. The enhancement of the mechanical properties of the coatings can be attributed to the synergistic effects of hydrogen bonding within the polyurethane matrix, the rigid benzene ring

structure of SBSi, and the covalent cross-linking network formed by SBSi. The findings of this study may provide inspiration for the design of subsequent Schiff base-modified polyurethanes and create new opportunities for their practical marine applications.

CRediT authorship contribution statement

Silei Liu: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Rongrong Chen:** Writing – review & editing, Funding acquisition. **Qi Liu:** Resources, Data curation. **Jingyuan Liu:** Resources, Data curation. **Jiahui Zhu:** Resources, Data curation. **Gaohui Sun:** Investigation. **Hongsen Zhang:** Writing – review & editing, Software. **Jun Wang:** Supervision, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109138>.

Data availability

Data will be made available on request.

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